PADHAKU
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CHAPTER 1

Patterns in Mathematics

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Figure it Out · 1.1 What is Mathematics?

1 **Can you think of other examples where mathematics helps us in our everyday lives?**

💡 **SAMPLE ANSWER**

- Shopping: adding prices and counting the change.
- Cooking: measuring ingredients and cooking time.
- Time: reading clocks and calendars, planning a timetable.
- Travel: finding distance, speed and bus/train timings.
- Home: finding the area of a room or plot, saving money in a budget.

2 **How has mathematics helped propel humanity forward? (Think of scientific experiments, economy and democracy, bridges and buildings, TVs, phones, computers, vehicles, calendars, clocks, etc.)**

💡 **SAMPLE ANSWER**

- Science: experiments use measurement and calculation; patterns in the motion of planets led to the theory of gravitation and to rockets to the Moon and Mars.
- Economy and democracy: banking, budgets, taxes and counting votes all use numbers.

- Buildings and bridges: engineers calculate lengths, angles and loads so structures are safe.
- Machines: phones, computers, TVs, cars, trains and planes are designed using mathematics.
- Calendars and clocks come from patterns in the motion of the sun, moon and earth.

Figure it Out · 1.2 Patterns in Numbers

1 **Can you recognise the pattern in each of the sequences in Table 1?**

✔ ANSWER

Yes. Each sequence follows a rule:

- All 1's: every number is 1.
- Counting numbers: add 1 each time.
- Odd numbers: start at 1, add 2 each time.
- Even numbers: start at 2, add 2 each time.
- Triangular numbers: add 2, then 3, then 4, ... (1, 1+2, 1+2+3, ...).
- Squares: 1×1, 2×2, 3×3, ...
- Cubes: 1×1×1, 2×2×2, 3×3×3, ...
- Virahānka numbers: each number = sum of the two before it (3 = 1+2, 5 = 2+3, 8 = 3+5).
- Powers of 2: multiply by 2 each time.
- Powers of 3: multiply by 3 each time.

2 **Rewrite each sequence of Table 1 with the next three numbers, and write the rule for each sequence.**

✔ ANSWER

Sequence	Next three numbers	Rule
All 1's: 1, 1, 1, ...	1, 1, 1	Always 1
Counting: ..., 6, 7	8, 9, 10	Add 1
Odd: ..., 11, 13	15, 17, 19	Add 2
Even: ..., 12, 14	16, 18, 20	Add 2
Triangular: ..., 21, 28	36, 45, 55	Add 8, then 9, then 10
Squares: ..., 36, 49	64, 81, 100	8×8, 9×9, 10×10
Cubes: ..., 125, 216	343, 512, 729	7×7×7, 8×8×8, 9×9×9
Virahānka: ..., 13, 21	34, 55, 89	Add the previous two numbers
Powers of 2: ..., 32, 64	128, 256, 512	Multiply by 2
Powers of 3: ..., 243, 729	2187, 6561, 19683	Multiply by 3

Figure it Out · 1.3 Visualising Number Sequences

1 **Copy the pictorial representations of the number sequences in Table 2 and draw the next picture for each sequence.**

💡 SAMPLE ANSWER

Next pictures (6th in each row):

- All 1's: 1 dot.
- Counting numbers: a row of 6 dots.
- Odd numbers: 11 dots in an L-shape (6 across and 6 down, sharing the corner).
- Even numbers: 12 dots in 2 rows of 6.
- Triangular numbers: 21 dots in a triangle with rows 1, 2, 3, 4, 5, 6.
- Squares: 36 dots in a 6 × 6 square.
- Cubes: 216 small cubes in a 6 × 6 × 6 cube.

2 **Why are 1, 3, 6, 10, 15, ... called triangular numbers? Why are 1, 4, 9, 16, 25, ... called squares? Why are 1, 8, 27, 64, 125, ... called cubes?**

✔ ANSWER

- **Triangular numbers:** these many dots can be arranged in a triangle (rows of 1, 2, 3, ...). E.g. 10 = 1 + 2 + 3 + 4.
- **Squares:** these many dots make a square with equal rows and columns. E.g. 16 = 4 × 4.

- **Cubes:** these many small cubes make a big cube of equal length, width and height. E.g. $27 = 3 \times 3 \times 3$.

3 **36 is both a triangular number and a square number. Make pictures showing this. Try representing some other numbers pictorially in different ways.**

💡 SAMPLE ANSWER

- **Triangle:** rows of 1, 2, 3, 4, 5, 6, 7, 8 dots → $1+2+3+4+5+6+7+8 = 36$.
- **Square:** 6 rows of 6 dots → $6 \times 6 = 36$.

Other examples:

- 1 is a triangle, a square and a cube (just 1 dot).
- 6: a triangle (1+2+3) and also a rectangle of 2×3.
- 16: a square of 4×4 and also a rectangle of 2×8.

4 **What would you call the sequence 1, 7, 19, 37 (dots in hexagon shapes)?**

They are called hexagonal numbers. Draw these. What is the next number in the sequence?

✔ ANSWER

Differences: $7 - 1 = 6$, $19 - 7 = 12$, $37 - 19 = 18$ (they go up by 6).

Next difference = 24.

37 + 24 = **61**

So the next hexagonal number is **61** (a hexagon with 5 dots on each side).

5 **Can you think of pictorial ways to visualise the sequence of Powers of 2? Powers of 3?**

💡 SAMPLE ANSWER

- **Powers of 2:** start with 1 dot. Each time, make 2 copies of the last picture and put them side by side: 1, 2, 4, 8, 16 dots.
- **Powers of 3:** start with 1 dot. Each time, make 3 copies of the last picture: 1 dot → 3 dots in a row → 3 rows of 3 = 9 dots (a square) → 3 such squares = 27 dots (a cube) → 81, ...

Another way: a tree where each branch splits into 2 (or 3) new branches at every step.

? 1.4 Relations among Number Sequences (in-text questions)

? **Adding odd numbers: 1 = 1, 1 + 3 = 4, 1 + 3 + 5 = 9, ... Why does this happen? Do you think it will happen forever?**

✔ ANSWER

Yes, it happens **forever**.

Reason: a square of dots can be grown into the next bigger square by adding an L-shaped border of dots. The borders have 1, 3, 5, 7, ... dots (the next odd number each time).

So adding the first n odd numbers always gives n × n.

? **How can we partition the dots in a square grid into odd numbers of dots: 1, 3, 5, 7, ...?**

✔ ANSWER

Split the square into L-shapes starting from one corner:

- the corner dot = 1
- the next L-shape = 3 dots
- the next L-shape = 5 dots, then 7, 9, 11, ...

For a 6 × 6 square: 1 + 3 + 5 + 7 + 9 + 11 = 36.

? **By drawing a similar picture, can you say what is the sum of the first 10 odd numbers?**

✔ ANSWER

The first 10 odd numbers fill a 10 × 10 square.

1 + 3 + 5 + ... + 19 = 10 × 10 = **100**

? **By imagining a similar picture, can you say what is the sum of the first 100 odd numbers?**

✔ ANSWER

The first 100 odd numbers fill a 100 × 100 square.

1 + 3 + 5 + ... + 199 = 100 × 100 = **10000**

? **Adding up and down (1, 1+2+1, 1+2+3+2+1, ...) gives square numbers. Can you find a similar pictorial explanation?**

✔ ANSWER

Yes. Draw a square of dots and count the dots along the slanting lines (diagonals).

In a 4 × 4 square the diagonals have 1, 2, 3, 4, 3, 2, 1 dots.

So 1 + 2 + 3 + 4 + 3 + 2 + 1 = 4 × 4 = 16. (See Figure it Out Q1 below.)

Figure it Out · 1.4 Relations among Number Sequences

1 **Can you find a similar pictorial explanation for why adding counting numbers up and down, i.e., 1, 1 + 2 + 1, 1 + 2 + 3 + 2 + 1, ..., gives square numbers?**

✔ ANSWER

Draw a square grid of dots and tilt it like a diamond (or look along its diagonals).

Count the dots row by row of the diamond:

- 2 × 2 square: 1, 2, 1 → 1 + 2 + 1 = 4
- 3 × 3 square: 1, 2, 3, 2, 1 → 9
- 4 × 4 square: 1, 2, 3, 4, 3, 2, 1 → 16

A square of side n always has diagonals 1, 2, ..., n, ..., 2, 1, so the sum is **n × n**.

2 **What will be the value of 1 + 2 + 3 + ... + 99 + 100 + 99 + ... + 3 + 2 + 1?**

✔ ANSWER

This is adding up to 100 and back down.

It equals a 100 × 100 square of dots.

Value = 100 × 100 = **10000**

3 **Which sequence do you get when you add the All 1's sequence up? What do you get when you add it up and down?**

✔ ANSWER

Up: 1, 1 + 1, 1 + 1 + 1, ... = 1, 2, 3, 4, ... → **counting numbers**.

Up and down: 1, 1 + 1 + 1, 1 + 1 + 1 + 1 + 1, ... = 1, 3, 5, 7, ... → **odd numbers**.

4 **Which sequence do you get when you start to add the counting numbers up? Can you give a smaller pictorial explanation?**

✔ ANSWER

1 = 1

1 + 2 = 3

1 + 2 + 3 = 6

1 + 2 + 3 + 4 = 10

We get **triangular numbers** 1, 3, 6, 10, 15, ...

Picture: put 1 dot in the first row, 2 in the second, 3 in the third, … The dots form a triangle (a right-angled staircase triangle).

5 **What happens when you add up pairs of consecutive triangular numbers (1 + 3, 3 + 6, 6 + 10, 10 + 15, …)? Which sequence do you get? Why? Explain with a picture.**

✔ ANSWER

1 + 3 = 4, 3 + 6 = 9, 6 + 10 = 16, 10 + 15 = 25

We get **square numbers**.

Why: draw a square of dots and cut it along a diagonal (staircase line). One part is a triangle of one size and the other part is the next triangle.

E.g. a 4 × 4 square = triangle of 6 dots (1+2+3) + triangle of 10 dots (1+2+3+4) = 16.

6 **Add up powers of 2 starting with 1: 1, 1 + 2, 1 + 2 + 4, 1 + 2 + 4 + 8, … Now add 1 to each of these numbers — what do you get? Why?**

✔ ANSWER

Sums: 1, 3, 7, 15, 31, …

Add 1: 2, 4, 8, 16, 32, … → **powers of 2**.

Why: 1 + 1 = 2; 2 + 2 = 4; 4 + 4 = 8; 8 + 8 = 16 …

Each time the extra 1 joins the sum to make the next power of 2 (doubling).

7 **What happens when you multiply the triangular numbers by 6 and add 1? Which sequence do you get? Explain with a picture.**

✔ ANSWER

$1 \times 6 + 1 = 7$

$3 \times 6 + 1 = 19$

$6 \times 6 + 1 = 37$

$10 \times 6 + 1 = 61$

We get **hexagonal numbers** 7, 19, 37, 61, ...

Picture: a hexagon of dots = 1 dot in the centre + 6 equal triangles of dots around it.

8 **Add up hexagonal numbers: 1, 1 + 7, 1 + 7 + 19, 1 + 7 + 19 + 37, ... Which sequence do you get? Can you explain it using a picture of a cube?**

✔ ANSWER

$1 = 1$

$1 + 7 = 8$

$1 + 7 + 19 = 27$

$1 + 7 + 19 + 37 = 64$

We get **cubes** 1, 8, 27, 64, ...

Picture: a cube seen from one corner looks like a hexagon. Build a cube by adding layers at a corner; each new layer (three faces together) has a hexagonal number of small cubes: 1, 7, 19, 37, ...

9 **Find your own patterns or relations in and among the sequences in Table 1. Can you explain why they happen?**

💡 SAMPLE ANSWER

Some examples:

- Odd + next odd = even numbers that are multiples of 4: 1 + 3 = 4, 3 + 5 = 8, 5 + 7 = 12.
- Each even number = 2 × counting number (2 = 2×1, 4 = 2×2, ...): picture of 2 equal rows.
- Adding consecutive counting numbers gives odd numbers: 1 + 2 = 3, 2 + 3 = 5, 3 + 4 = 7.
- 1 + 8 = 9 and 1 + 8 + 27 = 36: sums of cubes are squares of triangular numbers (1, 9, 36, 100).

🧮 Figure it Out · 1.5 Patterns in Shapes

1 **Can you recognise the pattern in each of the sequences in Table 3?**

✔ ANSWER

- **Regular polygons:** each shape has one more side than the last (3, 4, 5, ... 10 sides).
- **Complete graphs:** K2, K3, ... have 2, 3, 4, ... dots, and every dot is joined to every other dot.
- **Stacked squares:** squares of side 1, 2, 3, ... made of little squares.

- **Stacked triangles:** big triangles made of 1, 2, 3, … rows of little triangles.
- **Koch snowflake:** each line segment is replaced by a ‘speed bump’ to get the next shape.

2 **Redraw each sequence in Table 3. Can you draw the next shape in each sequence? Why or why not? Describe the rule for each sequence.**

💡 **SAMPLE ANSWER**

- **Regular polygons:** next is an 11-sided polygon (hendecagon). Rule: add one side.
- **Complete graphs:** next is K7: 7 dots, each joined to all the others (21 lines). Rule: add one dot and join it to every dot.
- **Stacked squares:** next is a 6 × 6 square (36 little squares). Rule: add one row and one column.
- **Stacked triangles:** next has 6 rows (36 little triangles). Rule: add one row at the bottom.
- **Koch snowflake:** the next shape is very hard to draw because the line segments become tiny and too many. Rule: replace every segment by a speed bump.

Figure it Out · 1.6 Relation to Number Sequences

1 **Count the number of sides in each shape of Regular Polygons. Which number sequence do you get? What about the number of corners? Do you get the same sequence? Why?**

✔ ANSWER

Sides: 3, 4, 5, 6, 7, 8, 9, 10 → **counting numbers starting from 3**.

Corners: 3, 4, 5, 6, 7, 8, 9, 10 → **the same sequence**.

Why: in a closed shape, each side ends at a corner where it meets the next side, so number of sides = number of corners.

2 **Count the number of lines in each shape of Complete Graphs. Which number sequence do you get? Why?**

✔ ANSWER

K2 = 1, K3 = 3, K4 = 6, K5 = 10, K6 = 15 lines.

This is the **triangular numbers**.

Why: when a new dot is added, it is joined to every old dot.

So we add 1, then 2, then 3, then 4, ...: 1, 1+2, 1+2+3, 1+2+3+4, ...

3 **How many little squares are there in each shape of Stacked Squares? Which number sequence? Why?**

✔ ANSWER

1, 4, 9, 16, 25 little squares → **square numbers**.

Why: each shape has equal rows and columns: 1×1, 2×2, 3×3, 4×4, 5×5.

4 **How many little triangles are there in each shape of Stacked Triangles? Which number sequence? Why? (Hint: how many triangles are there in each row?)**

✔ ANSWER

Rows have 1, 3, 5, 7, ... little triangles (odd numbers).

1 = 1

1 + 3 = 4

1 + 3 + 5 = 9

1 + 3 + 5 + 7 = 16

Total: 1, 4, 9, 16, 25 → **square numbers**, because adding odd numbers gives squares.

5 **In the Koch Snowflake, each line segment is replaced by a 'speed bump'. How many total line segments are there in each shape? What is the number sequence?**

✔ ANSWER

A speed bump has 4 small segments, so each step multiplies the number of segments by 4.

Shape 1: 3

Shape 2: 3 × 4 = 12

Shape 3: 12 × 4 = 48

Shape 4: 48 × 4 = 192

Sequence: **3, 12, 48, 192, 768, ...** (3 times powers of 4).

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**